

Exp 8

Code :

%Parameters

N=1000; %Number of samples for the signal

SNR_dB = -10; %Signal-to-Noise ratio in dB

threshold =1.5; %Detection threshold

PU_active = true; %Primary User active or not (1=active, 0=inactive)

%Generate primary user's signal if PU is active

if PU_active

PU_signal =sqrt(10^(SNR_dB/10))* randn(1,N);%Scaled random signal

else

PU_signal =zeros(1,N); %No signal if PU is inactive

end

%Add Gaussian noise to simulate the noisy environment

noise = randn (1,N);

%Received signal (signal+noise if PU is active, otherwise only noise)

received_signal=PU_signal-noise;

%Energy Detection:Calculate the energy of the received signal

energy=sum(abs(received_signal).^2)/N;

%Threshold Comparison:Decide if PU is active based on energy

if energy>threshold

disp('Channel is occupied by Primary User (PU)');

else

disp('Channel is free,Cognitive Radio can transmit');

end

%Plotting the received signal for visualization

figure;

plot(received_signal);

title('Received Signal');

xlabel('Sample Index');

ylabel('Amplitude');

grid on;

Output:

